

Copulas and financial applications (Suite 2)

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1 Financial applications in Credit portfolio modeling

5. Financial applications in Credit portfolio modeling

Objective: pricing & hedging of structured credit products.

m underlying defaultable securities, whose default times are $\tau_1, \dots, \tau_m$.

The payoff is $V_T := \psi(\tau_1, \dots, \tau_m)$. $\Rightarrow V_t = \mathbb{E}_Q[\psi(\tau_1, \dots, \tau_m)]$.

We need the joint law of $(\tau_1, \dots, \tau_m)$ under the risk-neutral probability Q !

To simplify: no interest rate risk, no recovery risk.

A source of inspiration: Merton's structural models

Every underlying i is related to an “asset value” process”
 $(X_{it})_{t \in [0, T]}$, the measures the “health/soundness” of i .

Standard assumption: $X_{it} = \rho_i W_t + \sqrt{1 - \rho_i^2} W_{it}$, for $m + 1$
independent Brownian motions (W_t) , (W_{it}) , $i \in \{1, \dots, m\}$.

A single systemic factor, m idiosyncratic factors.

When X_{it} falls below a threshold $\equiv$ default event of i (intuition)
...but **we rather postulate** $(\tau_i \leq T) = (X_{iT}/\sqrt{T} \leq b_{iT})$ for every T
 $\Rightarrow$ a sequence of q default barriers $(b_{i, T_k})_{k=1, \dots, q}$ for every name i .

Marginal laws: under the standard assumption, asset values are Gaussian, and

$$F_i(T) := \mathbb{Q}(\tau_i \leq T) = \mathbb{Q}(X_{iT}/\sqrt{T} \leq b_{iT}) = \Phi(b_{iT}).$$

The r.v. X_{iT} , $i \in \{1, \dots, m\}$ are conditionally independent, given W_T .

$$\begin{aligned} \mathbb{Q}(\tau_i \leq T | W_T/\sqrt{T} = w) &= \mathbb{Q}(X_{iT}/\sqrt{T} \leq b_{iT} | W_T/\sqrt{T} = w) \\ &= \Phi\left(\frac{b_{iT} - \rho_i w}{\sqrt{1 - \rho_i^2}}\right). \end{aligned}$$

Corporate credit risk: Credit-Default-Swaps quotes allow the fit of risk-neutral marginal laws, i.e. of the thresholds $(b_i, \tau_k)_{k=1, \dots, q}$ and every name.

Joint laws: a **naive assumption** is to consider the copula C between $(\tau_1, \dots, \tau_m)$, and to set

$$\mathbb{Q}(\tau_1 \leq T_1, \dots, \tau_m \leq T_m) = C(F_1(T_1), \dots, F_m(T_m)), \quad (1)$$

for **every time horizons** $T_1, \dots, T_m$.

Standard choice: C is a Gaussian copula, whose matrix of parameter Σ is “inspired” from the usual Merton’s model (single maturity), i.e.

$$\Sigma := [\rho_i \rho_j 1(i \neq j) + 1(i = j)]_{1 \leq i, j \leq m}.$$

For the payoff is $V_T = \tilde{\psi}(1(\tau_1 \leq T), \dots, 1(\tau_m \leq T))$, this is sufficient, because

$$\begin{aligned}\mathbb{Q}(\tau_1 \leq T, \dots, \tau_m \leq T) &= \mathbb{E}_Q \left[\mathbb{E}_Q [1(\tau_1 \leq T, \dots, \tau_m \leq T) | W_T] \right] \\ &= \int \prod_{i=1}^m \mathbb{Q}(X_i \leq b_{iT} \sqrt{T} | W_T / \sqrt{T} = w) \phi(w) dw \\ &= \int \prod_{i=1}^m \Phi\left(\frac{b_{iT} - \rho_i w}{\sqrt{1 - \rho_i^2}}\right) \phi(w) dw.\end{aligned}$$

To quote CDOs, most dealers assume $\rho_i = \rho_j =: \rho$, for every $(i, j) \in \{1, \dots, m\}^2 \Rightarrow$ a single **implied correlation** to price a complex CDOs with m underlings!

Credit portfolio modeling

Unfortunately, this way of working is not satisfying:

- (i) for different time horizons, different implied correlations, to recover market quotes (“correlation skew”) $\Rightarrow$ the Gaussian copula model of the law of $(\tau_1, \dots, \tau_m)$ is **inconsistent with the market**.
- (ii) some structured credit products necessitate the specification of $\mathbb{Q}(\tau_1 \leq T_1, \dots, \tau_m \leq T_m)$ for different vectors of time horizons $(T_1, \dots, T_m)$.

$\Rightarrow$ **Base correlation**, to solve (i), and to price standard CDOs.

Indeed, to price standard CDOs tranches, it is sufficient to evaluate

$$\mathbb{E}_{\mathbb{Q}} \left[(L_{T_k} - K)^+ \right], \text{ for different } K \text{ and } T_k \leq T, \text{ where}$$
$$L_{T_k} = \sum_{i=1}^m 1(\tau_i \leq T_k)(1 - R_i).$$

This is a model of dependencies between the binary events

$1(\tau_i \leq T_k)$, **for different i but for the same k !** A type of “incomplete model”.

To solve (ii), more sophisticated (dynamic) models are necessary...

Ex.: Forward starting CDOs, whose payoff depend on the joint law (L_{T_1}, L_{T_2}) .

In this case, the market information on standard CDO tranches (implied correlations) is most often insufficient for the sake of calibration of a model as (1).

An idea would be to directly postulate an increasing “Poisson-type” process (L_t) .

For the pricing/hedging of standard CDOs tranches, several copulas have been proposed, beside the Gaussian copula.

A single horizon T and m asset values $X_i := X_{iT}$ and $(\tau_i \leq T) = (X_i \leq b_i)$, $i \in \{1, \dots, m\}$.

$\Rightarrow$ postulate the joint law of $(X_1, \dots, X_m)$ is sufficient to calculate $\mathbb{E}[(L_T - K)^+]$, and the to get prices of standard CDO tranches.

Some CDS quotes yield the marginal laws (of τ_i or X_i , equivalently).

The remaining stone: the dependence structure between $(X_1, \dots, X_m)$.

(a). Postulate

$X_i := B_i(\rho V + \sqrt{1 - \rho^2} \varepsilon_i) + (1 - B_i)(\bar{\rho} V + \sqrt{1 - \bar{\rho}^2} \bar{\varepsilon}_i)$, where

- $B_i \in \{0, 1\} \sim \text{Bernoulli}(\rho)$,
- $\rho, \bar{\rho} \in (-1, 1)$,
- V, B_i, ε_i and $\bar{\varepsilon}_i$ are $\mathcal{N}(0, 1)$ and mutually independent.

$$\begin{aligned} \mathbb{Q}(\tau_i \leq T | V) &= \rho \Phi\left(\frac{\Phi^{-1}(F_i(T)) - \rho V}{\sqrt{1 - \rho^2}}\right) \\ &+ (1 - \rho) \Phi\left(\frac{\Phi^{-1}(F_i(T)) - \bar{\rho} V}{\sqrt{1 - \bar{\rho}^2}}\right), \end{aligned}$$

and $\mathbb{Q}(\tau_1 \leq T, \dots, \tau_m \leq T)$ is obtained by integration w.r.t. the law of the systemic factor V .

The underlying copula is a mixture of Gaussian copulas.

Denote by $Cop_{G,\theta}$ a bivariate Gaussian copula with correlation parameter θ .

$$Cop_{\tau_1, \tau_2} = p^2 Cop_{G, p^2} + \bar{p}^2 Cop_{G, \bar{p}^2} + 2p(1-p) Cop_{G, p\bar{p}},$$

with $\bar{p} := 1 - p$.

(b). Postulate

$$X_i := Y_i \sqrt{\nu} / \sqrt{Z}, Y_i = \rho Y + \sqrt{1 - \rho^2} \varepsilon_i, \text{ where}$$

- $Z \sim$ chi-2 law with ν degrees of freedom,
- $\rho \in (-1, 1)$,
- Y and the ε_i are $\mathcal{N}(0, 1)$ and mutually independent,
- $Z \perp (Y, \varepsilon_1, \dots, \varepsilon_m)$.

$$\Rightarrow (Y_1, \dots, Y_m) \sim \mathcal{N}(0_m, \Sigma), \Sigma := [1(i=j) + 1(i \neq j)\rho^2]_{1 \leq i, j \leq m}.$$

Given (Y, Z) , the default events are independent: [a two-factor model](#).

$$\mathbb{Q}(\tau_i \leq T | Y, Z) = \Phi\left(\frac{\sqrt{Z/\nu} T_\nu^{-1}(F_i(T)) - \rho Y}{\sqrt{1 - \rho^2}}\right).$$

The underlying copula is [Student](#) with parameters (ν, Σ) .

(c). Postulate a single systemic factor $V \sim \Gamma(1/\theta)$, $\theta > 0$, and that the conditional probability of default of i before T is

$$p_{i|V}(T) := \exp\left(V\{1 - F_i(T)^{-\theta}\}\right).$$

Equivalently, the asset value is $X_i := \psi(-\ln U_i/V)$, where $U_i \sim \mathcal{U}([0, 1])$, ψ is the Laplace transform of V and V and the U_i are mutually independent.

The underlying copula of $(X_1, \dots, X_m)$ is [Clayton](#), with a parameter θ .

(d). Postulate $X_i := \min(\tilde{X}_i, X)$, where

- $X \sim \mathcal{E}(1 - \alpha)$ and $\tilde{X}_i \sim \mathcal{E}(\alpha)$, $\alpha \in (0, 1)$;
- X and the $\tilde{X}_i$ are mutually independent.

$\Rightarrow X_i \sim \mathcal{E}(1)$.

The survival probability of i given X is

$$\mathbb{Q}(\tau_i > T | X) = 1 \left(X > -\ln(S_i(T)) \right) S_i(T)^\alpha,$$

where $S_i(T) = \mathbb{Q}(\tau_i > T) = 1 - F_i(T)$.

The underlying survival copula of $(X_1, \dots, X_m)$ is [Marshall-Olkin](#).

Once the latter models are calibrated to the most junior CDO tranches, they yield prices for the most senior tranches.

The performances in terms of fit are similar for most of these models, except Marshall-Olkin model: its allows a better fit of the super-senior tranche (sensitive to the risk of many defaults in the pool).

But the most challenging part is hedging!

Ref.: Burtschell, Gregory, Laurent (2008). A comparative analysis of CDO pricing models.

2 Copulas and time series

- The idea
- Copulae of serial dependence
- The Markov property
- Examples of stationary models

Goal: modeling of discrete/continuous time financial time series $(\mathbf{X}_i)_{t \in E}$, $\mathbf{X} \in \mathbb{R}^m$, with copulas (E denotes $\mathbb{Z}$ or $\mathbb{R}$).

Now: time dependencies, stationarity, Markovian (or not) features, etc.

No additional assumption, as martingale properties (typical of the “risk-neutral” world).

When $m > 1$, a complex task, in general (even if some solutions have been proposed in the literature).

When $m = 1$, the law of $(X_t)_{t \in \mathbb{Z}}$ is defined by the joint law of

$$\vec{X}_{\mathbf{t}} := (X_{t_1}, X_{t_2}, \dots, X_{t_q})$$

for every $\mathbf{t} := (t_1, \dots, t_q)$, $t_1 < t_2 < \dots < t_q$ and every $q \in \mathbb{N}$.

We define/estimate/simulate the process (X_t) by

- the specification/estimation of F_t , the **unconditional** law of X_t (margin) for every t ;
- and the specification/estimation of the copula of $\vec{X}_{\mathbf{t}}$, when $\mathbf{t} := (t_1, t_2, \dots, t_q)$. It is denoted $C_{\mathbf{t}}$.

If we assume that the process is p -Markov, we just need the joint laws of $\vec{X}_{\mathbf{t}}$, when $\mathbf{t} = (t-p, t-p+1, \dots, t)$.

$\Rightarrow$ the law of X_t given $(X_{t-1}, X_{t-2}, \dots, X_{t-p})$, i.e. Markov transition probabilities.

$\Rightarrow$ to generate a process, it is “necessary” to specify a sequence of copulas.

But these copulas **cannot be independently specified**:

$$C_{1,2,3}(1, u_2, u_3) = C_{2,3,4}(u_2, u_3, 1), \text{ e.g.}$$

$\Rightarrow$ we need **more structure**.

With one-order Markov processes, it is sufficient to select/specify the copula of (X_t, X_{t+1}) , but we may be interested in describing $C_{1,2,3,\dots,q,\dots}$

If stationarity is assumed: all margins are the same cdf F , and $C_{1,2,\dots,q} = C_{1+h,2+h,\dots,q+h}, \forall h > 0$.

p -Markov + stationarity $\Rightarrow$ a single $p + 1$ copula + the cdf F are sufficient!

Example: model of an asset return process $(X_t)_{t \in \mathbb{Z}}$, where

- at every t , some exogenous information is available: Z_t , that may be deterministic (intra-day patterns).
- F_t , the law of X_t , is $\mathcal{N}(\mu_t, \sigma_t^2)$, where μ_t and σ_t depend on Z_t .
- the copula of (X_t, X_{t+1}) is C_{θ_t} , and θ_t depends on Z_t .

$\Rightarrow$ an iterative formula

$$\mathbb{P}(X_{t+1} \leq y | X_t = x, Z_t) = \partial_1 C_{\theta_t}(F_t(x), F_{t+1}(y)), \forall x, y.$$

Typically, Z_t contains low-frequency factors, calendar time, switching-regimes, etc.

6.2. Copulae of serial dependence

More generally ($m > 1$): copulas provide a way of modelling multivariate processes.

Consider a process $(\mathbf{X}_t)_{t \in E}$ in $\mathbb{R}^m$, in discrete time ($E = \mathbb{Z}$) or continuous time ($E = \mathbb{R}$).

The law of the process is defined by

- 1 the marginal unconditional cdfs' F_{it} of X_{it} , for every $i = 1, \dots, m$ and $t \in E$.
- 2 the qm -dimensional copula $C_{t_1, \dots, t_q}$, associated to $(\mathbf{X}_{t_1}, \dots, \mathbf{X}_{t_q})$, for every q and every $t_1 < \dots < t_q$.

But there are relations between all these distributions:

$$C_{t_1, t_2, t_3}(\mathbf{u}, \mathbf{v}, 1) = C_{t_1, t_2}(\mathbf{u}, \mathbf{v}), \text{ etc.}$$

6.2. Copulae of serial dependence

“By coupling different marginal distributions with different copula functions, copula-based time series models are able to model the dependence structure and the marginal behavior of a time series separately, allowing for a wide variety of marginal behaviors (such as skewness and fat tailedness) and dependence properties (such as asymmetric dependence and positive tail dependence).”
(Chen & Fan, 2002, published in 2006)

6.2. Copulae of serial dependence

This is a rather too general framework: too many (high-dimensional) copulas to be specified...

To be realistic, we need more structure.

If strong stationarity is imposed and **unconditional** margins: for every $i = 1, \dots, m$ and t , $F_{it} \sim F_i$.

Actually, this was the framework proposed by Chen and Fan (2006).

6.3. The Markov property

Q: Consider a (large) family of copulas. Under which conditions is the process $(X_t)_{t \in E}$ **one-order markovian**?

When this is the case, only univariate unconditional margins and C_{t_1, t_2} are necessary, $t_1 < t_2$, to describe (simulate) the process.

Darsow, N'Guyen and Olsen (1992), in dimension one ($m = p = 1$) have solved the problem.

Ibragimov (2005) has solved the p -order Markov problem ($m = 1$, p arbitrary).

Let $\mathcal{C}$ be the set of copulas on $[0, 1]^2$.

Define the product of two copulas C and $\tilde{C}$: for $(u, v) \in [0, 1]^2$,

$$(C * \tilde{C})(u, v) = \int_0^1 \partial_2 C(u, x) \partial_1 \tilde{C}(x, v) dx,$$

$$\partial_j C(u_1, u_2) = \frac{\partial C}{\partial u_j}(u_1, u_2), \quad j = 1, 2.$$

$(u, v) \mapsto (C * \tilde{C})(u, v)$ is a copula.

$$C * (\tilde{C} * \overline{C}) = (C * \tilde{C}) * \overline{C},$$

$$C^\perp * C = C * C^\perp = C^\perp,$$

$$C^+ * C = C * C^+ = C.$$

Theorem 1

Let $(X_t)_{t \in E}$ be a real process. The following are equivalent:

- (i) the transition probabilities $\mathbb{P}(s, x, t, A) = \mathbb{P}(X_t \in A | X_s = x)$ satisfy the Chapman-Kolmogorov equations

$$\mathbb{P}(s, x, t, A) = \int \mathbb{P}(u, \xi, t, A) \mathbb{P}(s, x, u, d\xi),$$

for all Borel sets A , all $s < t$ in E , for all $u \in (s, t)$ and for almost every $x \in \mathbb{R}$.

- (ii) For every s, u, t satisfying $s < u < t$,

$$C_{st} = C_{su} * C_{ut}.$$

In particular, in a discrete time setting, assume $(\mathbf{X}_t)_{t \in \mathbb{Z}}$ is stationary.

The copula of (X_0, X_1) is $C_{0,1} := C_1 := C$. The copula of (X_0, X_k) is $C_{0,k} := C_k$.

$$C_{k+1}(u, v) = \int_0^1 \partial_2 C_k(u, x) \partial_1 C(x, v) dx$$

$\Rightarrow$ all the copulas may be expressed in terms of C .

Satisfaction of the Chapman-Kolmogorov equations is a necessary but not sufficient condition for a Markov process.

Let C be a m -copula and $\tilde{C}$ be a n -copula. Define

$$\begin{aligned} C \star \tilde{C}(u_1, \dots, u_{m+n-1}) \\ = \int_0^{u_m} \partial_m C(u_1, \dots, u_{m-1}, \xi) \partial_1 \tilde{C}(\xi, u_{m+1}, \dots, u_{m+n-1}) d\xi. \end{aligned}$$

$C \star \tilde{C}$ is a copula. The $\star$ -product is associative.

When $m = n = 2$, $C * \tilde{C}(u, v) = C \star \tilde{C}(u, 1, v)$.

Theorem 2

A real-valued process $(X_t)_{t \in E}$ is a Markov process *iff* for all integer n and all $t_1, \dots, t_n$ in E , satisfying $t_1 < t_2 < \dots < t_n$,

$$C_{t_1, \dots, t_n} = C_{t_1, t_2} \star \dots \star C_{t_{n-1}, t_n},$$

where $C_{t_1, \dots, t_n}$ denotes the copula of $(X_{t_1}, \dots, X_{t_n})$.

Note that we have not imposed (weak or strong) stationarity.

Example: Brownian motions with non-Gaussian marginals.

For the usual Brownian motion (W_t) , the transition probabilities are

$$\mathbb{P}(s, x, t,] - \infty, y]) = \frac{1}{\sqrt{t-s}} \int_{-\infty}^y \phi\left(\frac{u-x}{\sqrt{t-s}}\right) du.$$

Marginals are Gaussian and time-dependent:

$$X_s \sim \mathcal{N}(0, s), \quad F_s(x) = \frac{1}{\sqrt{s}} \Phi\left(\frac{x}{\sqrt{s}}\right).$$

Since $\mathbb{P}(s, x, t,] - \infty, y]) = \partial_1 C_{st}(F_s(x), F_t(y))$, the associated copulas are

$$C_{st}(u, v) = \int_0^u \Phi \left(\frac{\sqrt{t}\Phi^{-1}(v) - \sqrt{s}\Phi^{-1}(x)}{\sqrt{t-s}} \right) dx.$$

And any non-Gaussian univariate cdf can replace the F_s .

6.4. Examples of stationary models

In this subsection, assume

- ① $m = 1$
- ② $(\mathbf{X}_t)_{t \in \mathbb{Z}} = (X_t)_{t \in \mathbb{Z}}$ is strongly stationary, and
- ③ the unconditional marginal cdf F_0 is unspecified

$\Rightarrow$ semiparametric copula-based one order Markov processes.

This is a usual situation in statistics/econometrics.

6.4. Examples of stationary models

Example: semiparametric regression transformation models

$$\Phi^{-1}(F_0(X_t)) = \alpha \cdot \Phi^{-1}(F_0(X_{t-1})) + \varepsilon_t,$$

where $\varepsilon_t \sim \mathcal{N}(0, 1 - \alpha^2)$, (ε_t) iid and ε_t independent of X_{t-1} .

$\Rightarrow$ AR time series generated by Gaussian copulas...

...but with non-normal marginal laws F_0 in general (Student, fat-tailed, etc.).

6.4. Examples of stationary models

Generalization:

$$\Lambda_1(X_t) = \Lambda_2(X_{t-1}) + \sigma(X_{t-1})\varepsilon_t,$$

for some monotonic transform Λ_1, Λ_2 arbitrary, $\inf_x \sigma(x) > 0$, (ε_t) iid and ε_t independent of X_{t-1} .

Close to GARCH-type ideas... but a bit too general (identifiability issues).

6.4. Examples of stationary models

But we can remove the independence restriction between the error term and past values.

Generalized semiparametric regression transformation models:

$$\Lambda_{1,\theta_1}(F_0(X_t)) = \Lambda_{2,\theta_2}(F_0(X_{t-1})) + \varepsilon_t, \quad E[\varepsilon_t | X_{t-1}] = 0,$$

where Λ_{1,θ_1} is increasing, F_0 is unknown and

$$\Lambda_{2,\theta_2}(v) = E[\Lambda_{1,\theta_1}(F_0(X_t)) | F_0(X_{t-1}) = v] = \int_0^1 \Lambda_{1,\theta_1}(u) c_\alpha(u, v) du.$$

$\Rightarrow \Lambda_2$ is entirely given by Λ_1 and the underlying copula!

6.4. Examples of stationary models

Generalized semiparametric regression transformation models:

When $\Lambda_1(v) = v$,

$$F_0(X_t) = \Lambda_{2,\theta}(F_0(X_{t-1})) + \varepsilon_t, \quad E[\varepsilon_t | X_{t-1}] = 0,$$

$$\Lambda_{2,\theta}(v) = E[U_t | U_{t-1} = v] = \int_0^1 u c_\alpha(u, v) du.$$

Simple explicit expressions are available for

Fairly-Gumbel-Morgenstern $C(u, v) = uv[1 + \alpha(1 - u)(1 - v)]$,
 $|\alpha| \leq 1$, Plackett copulas, etc.

6.4. Examples of stationary models

Quantile regression models:

$$F_0(X_t) = Q_q(F_0(X_{t-1}), \alpha) + \eta_t, \quad \mathbb{P}(\eta_t < 0 | X_{t-1}) = q \in (0, 1),$$

where $Q_q(U_{t-1}, \alpha)$ is the q -th conditional quantile of $U_t := F_0(X_t)$ given U_{t-1} :

$$\int_0^{Q_q(U_{t-1}, \alpha)} c(U_{t-1}, u, \alpha) du = q, \text{ for all } X_{t-1}.$$

This means $Q_q(U_{t-1}, \alpha) = C_{2|1}^{-1}(q|U_{t-1}, \alpha)$, where $C_{2|1}(\cdot|U_{t-1}, \alpha)$ is the conditional law of U_t given U_{t-1} .

See Bouyé and Salmon (2002) for explicit formulas.

3 Conditional copulas

- The challenge
- The MGARCH framework

7.1. The challenge

With **independent** random vectors in $\mathbb{R}^m$, the joint law of a sample $\mathcal{X}_n := (\mathbf{X}_1, \dots, \mathbf{X}_n)$, $\mathbf{X}_i \in \mathbb{R}^m$, is given by

- the marginal cdfs' $F_1, \dots, F_m$, and
- the underlying copula of $\mathbf{X}$.

$\Rightarrow$ inference, MLE, GOF, ... “more or less” as usual.

With **dependent** random vectors in $\mathbb{R}^m$, the joint law of a sample $\mathcal{X}_T := (\mathbf{X}_1, \dots, \mathbf{X}_T)$, $\mathbf{X}_t \in \mathbb{R}^m$, is given by

$$\mathcal{L}(\mathcal{X}_T) = \mathcal{L}(\mathbf{X}_1) \mathcal{L}(\mathbf{X}_2 | \mathbf{X}_1) \cdots \mathcal{L}(\mathbf{X}_T | \mathbf{X}_1, \dots, \mathbf{X}_{T-1}).$$

$\Rightarrow$ we need to describe $\mathcal{L}(\mathbf{X}_t | \mathbf{X}_1, \dots, \mathbf{X}_{t-1})$ for every $t = 1, \dots, T$.
Without any additional structure, this can be done by several ways, with copula-based ideas.

7.1. The challenge

a. Unconditional margins.

Assume an underlying stationary process $(\mathbf{X}_t)_{t \in \mathbb{Z}}$: F_k the unconditional cdf of X_{kt} , $k \in \{1, \dots, d\}$, any t .

$$\begin{aligned} \mathbb{P}(X_{1,t} \leq x_1, \dots, X_{m,t} \leq x_m | \mathbf{X}_1^{t-1}) \\ = \mathbb{P}(U_{1,t} \leq F_1(x_1), \dots, U_{d,t} \leq F_d(x_d) | \mathbf{X}_1^{t-1}), \end{aligned}$$

where $\mathbf{X}_t^{t'} = (\mathbf{X}_t, \mathbf{X}_{t+1}, \dots, \mathbf{X}_{t'})$, $t \leq t'$, and

$$\mathbf{U}_t := (U_{1,t}, \dots, U_{m,t}) := (F_1(X_{1,t}), \dots, F_m(X_{m,t})).$$

But the joint law of $(U_{1,t}, \dots, U_{d,t})$ given $\mathbf{X}_1^{t-1}$ is not a copula (no uniform margins)! And it has to be specified.

Do we really gain something with the switch $X_{k,t} \rightsquigarrow U_{k,t}$?

The first examples of “conditional” copula models in finance: Jondeau and Rockinger (2002, published in 2006)

The goal is to identify time-varying dependencies among asset returns.

- (z_{1t}, z_{2t}) standardized returns, deduced from GARCH(1,1)-type models: for $k = 1, 2$,

$$r_{kt} = \mu_{kt} + \varepsilon_{kt}, \quad \varepsilon_{kt} := \sigma_{kt} z_{kt},$$

$$\sigma_{kt}^2 = a_k + b_k \sigma_{k,t-1}^2 + c_k^+ (\varepsilon_{kt}^+)^2 + c_k^- (\varepsilon_{kt}^-)^2.$$

- $z_{kt} \sim F_k(\cdot, \theta_k)$, $k = 1, 2$ skewed Student-t (no time-varying parameters for margins)

What is the dependence structure of (z_{1t}, z_{2t}) ?

- a Gaussian copula with a time-varying parameter

① $\rho_t = \sum_{j=1}^M d_j 1\{(z_{1t}, z_{2t}) \in \mathcal{A}_j\},$
for some partition $(\mathcal{A}_1, \dots, \mathcal{A}_M)$, and $d_j \in (-1, 1),$
 $j = 1, \dots, M.$

- ② Alternatively,

$$\rho_t = (1 - \alpha - \beta)\bar{\rho} + \alpha\zeta_t + \beta\rho_{t-1},$$

$\zeta_t :=$ (rolling window) empirical correlation of $(z_{1t}, z_{2t}).$

- a Student copula with time-varying parameters that are driven by a hidden (homogenous) Markov chain (S_t) .

$$\rho_t = \bar{\rho}_0 S_t + \bar{\rho}_1 (1 - S_t),$$

$$\nu_t = \bar{\nu}_0 S_t + \bar{\nu}_1 (1 - S_t).$$

- Application to several couples of stock indices.

7.1. The challenge

b. **Conditional margins**, with the same conditioning variables.

This is the **conditional copula** approach (Patton, 2002) .

$$\begin{aligned} & \mathbb{P}(X_{1,t} \leq x_1, \dots, X_{m,t} \leq x_m \mid \mathbf{X}_1^{t-1}) \\ &= C(\mathbb{P}(X_{1,t} \leq x_1 \mid \mathbf{X}_1^{t-1}), \dots, \mathbb{P}(X_{m,t} \leq x_m \mid \mathbf{X}_1^{t-1}) \mid \mathbf{X}_1^{t-1}) \end{aligned}$$

Here, the split is between conditional marginal cdfs' and conditional laws of $(F_{1t}(X_{1t}), \dots, F_{mt}(X_{mt}))$.

This is more inline with copula modelling with independent observations.

But are these conditioning events always relevant **for any margin** ?

7.1. The challenge

Theorem 3 (Sklar's theorem for conditional copulas)

Let $\mathbf{X} \in \mathbb{R}^m$ and some sub- σ -algebra $\mathcal{B} \subset \mathcal{A}$. There exists a random map $C : [0, 1]^m \times \Omega \rightarrow [0, 1]$ s.t.

$$\begin{aligned}\mathbb{P}(\mathbf{X} \leq \mathbf{x} | \mathcal{B})(\omega) \\ &= C(\mathbb{P}(X_1 \leq x_1 | \mathcal{B})(\omega), \dots, \mathbb{P}(X_m \leq x_m | \mathcal{B})(\omega), \omega | \mathcal{B}) \\ &=: C(\mathbb{P}(X_1 \leq x_1 | \mathcal{B}), \dots, \mathbb{P}(X_m \leq x_m | \mathcal{B}) | \mathcal{B})(\omega),\end{aligned}$$

for every $\mathbf{x} \in \mathbb{R}^m$ and almost $\omega \in \Omega$. This map is $\mathcal{B}([0, 1]^m) \otimes \mathcal{B}$ measurable. For almost every $\omega \in \Omega$, $C(\cdot, \omega)$ is a copula and it is uniquely defined on the product of the values taken by $x_k \mapsto \mathbb{P}(X_k \leq x_k | \mathcal{B})(\omega)$, $k \in \{1, \dots, m\}$

C is called a **conditional copula** (Patton, 2001)

7.1. The challenge

Considering a filtration $(\mathcal{F}_t)_{t \geq 0}$, conditional copulas given $\mathcal{F}_t$ are random maps

$$C : [0, 1]^m \times \Omega \rightarrow [0, 1],$$

s.t. $C(\cdot|\omega)$ is a copula for every $\omega \in \mathcal{F}_t$.

For every conditional distribution given $\mathcal{F}_{t-1}$ there exist conditional copulas s.t.

$$\begin{aligned} C(\mathbb{P}(X_{1,t} \leq x_1 | \mathcal{F}_{t-1}), \dots, \mathbb{P}(X_{m,t} \leq x_m | \mathcal{F}_{t-1}) | \mathcal{F}_{t-1}) \\ = \mathbb{P}(\mathbf{X}_t \leq \mathbf{x} | \mathcal{F}_{t-1}), \end{aligned}$$

for every t and $\mathbf{x} = (x_1, \dots, x_m)$.

Extension of Sklar's theorem: existence and unicity of conditional copulas.

Note that the filtration is **the same** for the copula and **all** the margins. Moreover, it can include **exogenous variables** (covariates).

7.1. The challenge

Sklar's th. for conditional copulas $\Rightarrow$ a family of cond. copulas + "fully" conditional margins $\simeq$ the whole multivariate process.

In practice, how can we specify $P(X_{k,t} \leq x_k | \mathcal{F}_{t-1})$? Not so easy...

Implicitly, the underlying assumption is often

$$P(X_{k,t} \leq x_k | \mathcal{F}_{t-1}) = P(X_{k,t} \leq x_k | X_{k,t-1}), \forall k, t,$$

but not always!

7.1. The challenge

Jondeau and Rockinger's model was already based on the idea of conditional copulas:

Knowing $\mathcal{F}_{t-1}$, the copula of (z_{1t}, z_{2t}) = the copula of (r_{1t}, r_{2t}) , and it is a conditional copula!

- If we consider this as a model for $(z_{1t}, z_{2t}) \rightarrow$ unconditional margins (the law of z_{kt} does not depend on the past, contrary to the joint law of (z_{1t}, z_{2t})).
- If we consider this as a model for $(r_{1t}, r_{2t}) \rightarrow$ conditional margins.

7.1. The challenge

Generally speaking, with unconditional margins, the effort for modelling is rather unbalanced: it is a lot easier to model fixed unconditional margins than time-varying copulas.

Is it corresponding really to the **spirit of copula modelling** ?

Our goal is rather to put in the model knowledge about marginal conditional laws.

In other words, in general, unconditional margins **do not provide** a sufficient amount of information for the first stage of the modelling process.

Q: are the FX rate dependencies symmetric ?

- (z_{1t}, z_{2t}) standardized returns, deduced from GARCH(1,1)-type models: for $k = 1, 2$,

$$r_{kt} = \mu_{kt} + \varepsilon_{kt}, \quad \varepsilon_{kt} := \sigma_{kt} z_{kt},$$

$$\sigma_{kt}^2 = a_k + b_k \sigma_{k,t-1}^2 + c_k \varepsilon_{kt}^2.$$

- $z_{kt} \sim F_k(\cdot, \theta_k)$, $k = 1, 2$, Student-t (no time-varying parameters for margins !)
- a Gaussian copula with a time-varying parameter

$$\rho_t = \Lambda(\omega + \beta \rho_{t-1} + \alpha \sum_{j=1}^m \Phi^{-1}(u_{t-j}) \Phi^{-1}(v_{t-j})),$$

$$\Lambda(x) = (1 - \exp(-x)) / (1 + \exp(x)), \quad u_t = F_1(z_{1t}), v_t = F_2(z_{2t}).$$

- a Joe-Clayton copula (BB7)

$$C(u, v) = 1 - \left(1 - \{ [1 - (1 - u)^\kappa]^{-\gamma} + [1 - (1 - v)^\kappa]^{-\gamma} - 1 \}^{-1/\gamma} \right)^{1/\kappa},$$

$$\kappa := 1/\ln_2(2 - \lambda^U), \quad \gamma = -1/\ln_2(\lambda^L),$$

λ^U and λ^L : the upper and lower tail dependence parameters.

- alternatively:

$$\lambda_t^U = \Gamma \left(\omega + \beta \lambda_{t-1}^U + \alpha \sum_{j=1}^m |u_{t-j} - v_{t-j}| \right)$$

$$\Gamma(x) = (1 + \exp(-x))^{-1}, \text{ and similarly for } \lambda^L.$$

+ application to USD-Yen and USD-German mark. There are asymmetries !

7.2. The MGARCH framework

A usual way of enriching dependencies in econometrics, popularized by Chen and Fan (2006), Patton (2006).

For every t and $i = 1, \dots, m$,

$$X_{i,t} = \mu_i(\mathbf{Z}_{t-1}, \theta_i^{(1)}) + \sigma_i(\mathbf{Z}_{t-1}, \theta_i^{(2)})\varepsilon_{i,t},$$

$\mathbf{Z}_{t-1}$ measurable wrt $\mathcal{F}_{t-1} = \sigma(\mathbf{X}_{t-1}, \mathbf{X}_{t-2}, \dots, \mathbf{W}_{t-1}, \mathbf{W}_{t-2}, \dots)$,

$$\varepsilon_{i,t} | \mathcal{F}_{t-1} \sim F_{i,t},$$

$$\vec{\varepsilon}_t := (\varepsilon_{1,t}, \dots, \varepsilon_{m,t}) | \mathcal{F}_{t-1} \sim C_t(F_{1,t}, \dots, F_{m,t}),$$

Note the copula C_t and the margins F_{it} depend (or not!) on $\mathcal{F}_t$.

First stage: estimation of the $\theta_i^{(1)}$

For instance, for every i ,

$$\hat{\theta}_i^{(1)} := \arg \max_{\theta_i} \sum_{t=1}^T (X_{i,t} - \mu_i(\mathbf{Z}_{t-1}, \theta_i))^2 .$$

If the $\theta_i^{(1)}$ share some components, concatenate them into $\theta^{(1)}$, and set

$$\hat{\theta}^{(1)} := \arg \max_{\theta} \sum_{t=1}^T [\mathbf{X}_t - \mu(\mathbf{Z}_{t-1}, \theta)]' [\mathbf{X}_t - \mu(\mathbf{Z}_{t-1}, \theta)] .$$

Second stage: estimation of the $\theta_i^{(2)}$

For instance, the usual Gaussian QMLE: individually,

$$\hat{\theta}_i^{(2)} := \arg \min_{\theta_i} \sum_{t=1}^T \frac{(X_{i,t} - \mu_i(\mathbf{Z}_{t-1}, \hat{\theta}_i^{(1)}))^2}{\sigma_i^2(\mathbf{Z}_{t-1}, \theta_i)} + 2 \ln \sigma_i(\mathbf{Z}_{t-1}, \theta_i),$$

or jointly, if the $\theta_i^{(2)}$ share some components:

$$\hat{\theta}^{(2)} := \arg \min_{\theta} \sum_{i=1}^m \sum_{t=1}^T \frac{(\mathbf{X}_t - \mu_i(\mathbf{Z}_{t-1}, \hat{\theta}_i^{(1)}))^2}{\sigma_i^2(\mathbf{Z}_{t-1}, \theta)} + 2 \ln \sigma_i(\mathbf{Z}_{t-1}, \theta).$$

Note this is not the usual QMLE for MGARCH (no correlation in the criterion above).

Third stage: estimation of the joint law of $\vec{\varepsilon}_t$, given $\mathcal{F}_{t-1}$

A lot of liberty to specify C_t and the $F_{i,t}$.

Assume the **simplest** assumption: $F_{i,t} = F_i$, for every t , and $C_t(\mathbf{u}|\mathcal{F}_{t-1}) = C(\mathbf{u})$ for every $\mathbf{u} \in [0, 1]^d$.

Then, C is a "usual" **deterministic** copula of $\vec{\varepsilon}_t$ (the "**simplifying assumption**").

But C remains the **conditional** copula of $\mathbf{X}_t$ knowing $\mathcal{F}_{t-1}$ too:

$$\begin{aligned} \mathbb{P}(X_{1,t} \leq x_1, \dots, X_{m,t} \leq x_m | \mathcal{F}_{t-1}) \\ = C(\mathbb{P}(X_{1,t} \leq x_1 | \mathcal{F}_{t-1}), \dots, \mathbb{P}(X_{m,t} \leq x_m | \mathcal{F}_{t-1})) \end{aligned}$$

Indeed, given $\mathcal{F}_{t-1}$, $\varepsilon_{i,t}$ is a monotonic increasing function of $X_{i,t}$.

⇒ Assume C belongs to a parametric copula family

$$\mathcal{C} := \{C_\alpha, \alpha \in B \subset \mathbb{R}^q\}.$$

Evaluate the unobservable vectors $\vec{\varepsilon}_t$ by some pseudo-observations

$$\hat{\varepsilon}_{it} := \frac{X_{it} - \mu_i(\mathbf{Z}_{t-1}, \hat{\theta}^{(1)})}{\sigma_i(\mathbf{Z}_{t-1}, \hat{\theta}^{(2)})},$$

$$\hat{\alpha} := \arg \max_{\alpha} \sum_{t=1}^T \ln c_{\alpha} \left(\hat{F}_{1,T}(\hat{\varepsilon}_{1,t}), \dots, \hat{F}_{m,T}(\hat{\varepsilon}_{m,t}) \right),$$

$$\hat{F}_{i,T}(y) := T^{-1} \sum_{t=1}^T 1(\hat{\varepsilon}_{it} \leq y), \quad i = 1, \dots, m.$$

Consistency and asymptotic normality of $\hat{\alpha}$ in Chen and Fan (2002), Chen et al. (2009).

The estimated parameters from the conditional mean and variance **do not affect** the asymptotic distribution of $\hat{\alpha}$! This is a nice property.

In particular, we can use the usual parametric bootstrap methods to evaluate the limiting behavior of $\hat{\alpha}$.

Rémillard (2010) pointed out that the previous property is no longer true in a more general model, with volatility “contamination” accross asset returns:

$$\begin{aligned}\mathbf{X}_t &= \mu(\mathbf{Z}_{t-1}, \theta^{(1)}) + \Sigma(\mathbf{Z}_{t-1}, \theta^{(2)})\vec{\varepsilon}_t, \\ \mathcal{F}_{t-1} &= \sigma(\mathbf{X}_{t-1}, \mathbf{X}_{t-2}, \dots, \mathbf{Z}_{t-1}, \mathbf{Z}_{t-2}, \dots),\end{aligned}$$

The observations $(\vec{\varepsilon}_t)$ are iid, $\vec{\varepsilon} \sim C$.

But the matrix Σ_t may be non-diagonal.

4 Pair-copula constructions

8. Pair-copula constructions

A drawback of copula models, especially in high dimension ($m \gg 1$): not easy to exhibit “the perfect” copula (family) a priori.

And the choice is relatively reduced.

Idea: define highly flexible m -dimensional laws with the help of bivariate copulas and marginal law only.

8. Pair-copula constructions

Illustration: when $m = 2$, let $\mathbf{X} = (X_1, X_2) \sim (C_{12}, F_1, F_2)$.

$$\begin{aligned}f(x_1, x_2) &= c_{12}(F_1(x_1), F_2(x_2))f_1(x_1)f_2(x_2) \\f_{1|2}(x_1|x_2) &= c_{12}(F_1(x_1), F_2(x_2))f_1(x_1)\end{aligned}\tag{2}$$

When $m = 3$, let $\mathbf{X} = (X_1, X_2, X_3) \sim (C_{123}, F_1, F_2, F_3)$.

$$f(x_1, x_2, x_3) = f_{1|23}(x_1|x_2, x_3)f_{23}(x_2, x_3)$$

Applying (2) given $X_3 = x_3$,

$$f_{1|23}(x_1|x_2, x_3) = c_{12|3}(F_1(x_1|x_3), F_2(x_2|x_3)|x_3)f_{1|3}(x_1|x_3)$$

8. Pair-copula constructions

$$\Rightarrow f(x_1, x_2, x_3) = c_{12|3}(F_1(x_1|x_3), F_2(x_2|x_3) | x_3) \\ c_{13}(F_1(x_1), F_3(x_3)) c_{23}(F_2(x_2), F_3(x_3)) f_1(x_1) f_2(x_2) f_3(x_3).$$

$\Rightarrow$ the law of (X_1, X_2, X_3) is defined by

- three margins F_k , $k = 1, 2, 3$;
- two bivariate copulas C_{13} and C_{23} ;
- a conditional copula $C_{12|3}$.

Different ways of decomposing the joint law of $\mathbf{X}$.

8. Pair-copula constructions

A commonly accepted assumption:

Conditional copulas are assumed to be “usual” copulas.

It is called the “simplifying” assumption (SA).

In the latter example, this means: for every x_1, x_2, x_3 ,

$$c_{12|3}(F_1(x_1|x_3), F_2(x_2|x_3)|x_3) = c_{12|3}(F_1(x_1|x_3), F_2(x_2|x_3))$$

Under (SA), the effect of conditioning variables is transferred only through conditional margins.

Pros: simpler for model specification and inference

Cons: may be too strong...

Vines = a graphical way of representing such decompositions into bivariate copulas.

Definition 1

$T = (N, E)$ is a *tree* with nodes N and edges E if E is a subset of unordered pairs of N with no cycle and there is a path between each pair of nodes. That is, there does not exist a sequence $a_1, \dots, a_k$ for $k > 2$ of elements of N such that

$$(a_1, a_2) \in E, \dots, (a_{k-1}, a_k) \in E, (a_k, a_1) \in E.$$

And for any $a, b \in E$, there exists a sequence $c_2, \dots, c_{k-1}$ of elements of N such that

$$(a, c_2) \in E, (c_2, c_3) \in E, \dots, (c_{k-1}, b) \in E.$$

T is a *connected tree* if there is a path between all nodes of T .

Vines are undirected graphs that allow nice specifications of multivariate joint distributions (copulas), and/or correlation matrices.

Definition 2

$\mathcal{V}(m)$ is a labeled *regular vine* on n elements if:

- ① $\mathcal{V}(m) = (T_1, T_2, \dots, T_{m-1})$.
- ② T_1 is a connected tree with nodes $N_1 = \{1, 2, \dots, m\}$ and edges E_1 . For $i = 2, \dots, m-1$, T_i is a connected tree with nodes $N_i = E_{i-1}$, and the cardinality of N_i is $m - i + 1$.
- ③ If a and b are nodes of T_i connected by an edge in T_i , where $a = \{a_1, a_2\}$ and $b = \{b_1, b_2\}$ (a_1, a_2, b_1 and b_2 are nodes of T_{i-1}), then exactly one of the a_i equals one of the b_i . This is the *proximity* condition: two nodes in tree i are only connected by an edge if these nodes share a common element in tree $i-1$.

An edge in tree T_j is an unordered pair of nodes of T_j , or equivalently, an unordered pair of edges of T_{j-1} .

We will consider only regular vines.

There are $m(m-1)/2$ edges in a regular vine on m variables.

A regular vine is called a *Canonical vine (C-vine)* if each tree T_i has a unique node of degree $m-i$, i.e. a node with maximum degree (i.e. number of connected neighbors).

A regular vine is called a *D-vine* if all nodes in T_1 have degree not higher than 2.

Fig. V1: C-vine with 5 variables and 10 edges

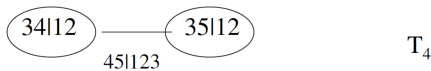
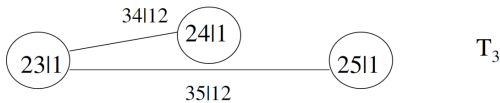
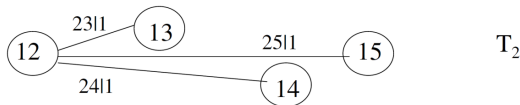
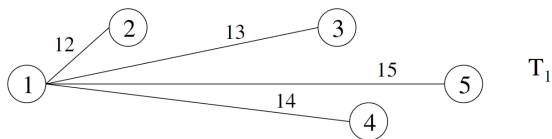


Fig. V2: D-vine with 5 variables and 10 edges

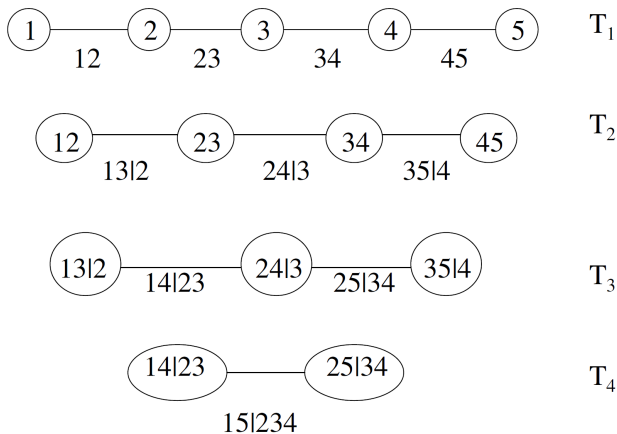


Fig. V3: R-vine with 5 variables and 10 edges

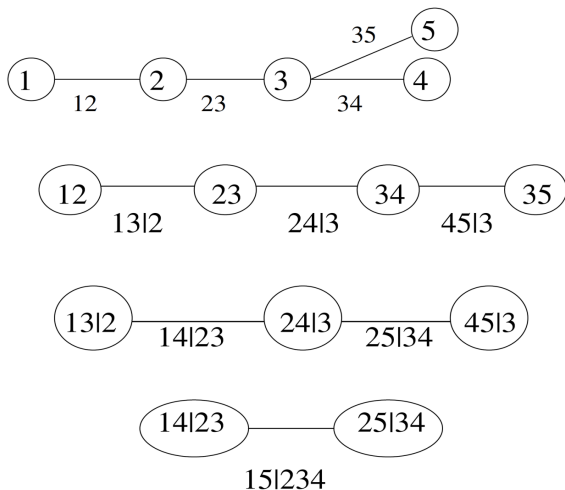
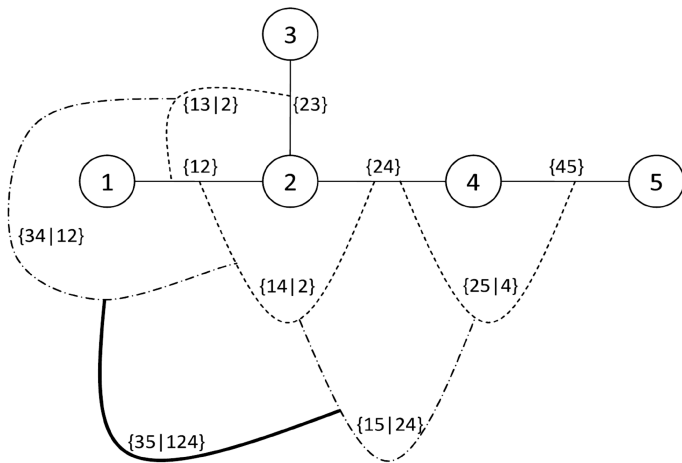


Fig. V4: R-vine with 5 variables and 10 edges



Definition 3

For $e \in E_i$, $i \leq m - 1$, the *constraint set* associated with e is the complete union of the elements in $\{1, \dots, m\}$ that are reachable from e by the membership relation. It is denoted by $V(e)$.

For $i = 1, \dots, m - 1$, any edge e in tree T_i will be denoted $j(e), k(e) | D(e)$, where

- $D(e) \subset \{1, \dots, m\}$ is the conditioning subset of e :

$$D(e) = V(j(e)) \cap V(k(e))$$

- $\{j(e), k(e)\} \subset \{1, \dots, m\}^2$ is the conditioned set of e :

$$\{j(e), k(e)\} = \{V(j(e)) \setminus D(e), V(k(e)) \setminus D(e)\}.$$

If $e \in E_i$, then $\#V(e) = i + 1$, $\#D(e) = i - 1$ and there are necessarily two elements in its conditioned set: see Bedford and Cooke (2002).

Any pair-copula decomposition can be described by a vine: the edges of the tree = indices used for the (conditional) bivariate copulas.

Any edge e in tree T_i will be built from two edges of T_{i-1} , which have a common node in T_{i-1} .

Denote these two edges $a := j(a), k(a)|D(a)$ and $b := j(b), k(b)|D(b)$.

Set $V(a) := \{j(a), k(a), D(a)\}$ and $V(b) := \{j(b), k(b), D(b)\}$.

The nodes a and b in T_i are joined by edge $e = j(e), k(e)|D(e)$ where

- (i) $D(e) = V(a) \cap V(b)$,
- (ii) $j(e) = \min \{i : i \in (V(a) \cup V(b)) \setminus D(e)\}$,
- (iii) $k(e) = \max \{i : i \in (V(a) \cup V(b)) \setminus D(e)\}$.

a and b can be connected in a regular vine only when the cardinal of $\{i : i \in (V(a) \cup V(b)) \setminus D(e)\}$ is two!

Example: in Figure V4, let us analyze the edge $e = 15|24$.

$$j(e) = 1, k(e) = 5, D(e) = \{2, 4\}.$$

$$a = 14|2, b = 25|4,$$

$$V(a) = \{1, 2, 4\}, V(b) = \{2, 5, 4\},$$

$$D(e) = V(a) \cap V(b) = \{2, 4\}$$

To build up a model on a regular vine tree with nodes $\mathcal{N} = \{N_1, \dots, N_{m-1}\}$ and edge set $\mathcal{E} := \{E_1, \dots, E_{m-1}\}$, one associates each edge $e = j(e), k(e) | D(e)$ with a bivariate conditional copula $C_{j(e), k(e) | D(e)}$.

$\mathbf{X}_{D(e)}$ denotes the sub-vector of $\mathbf{X}$ whose components belong to $D(e)$.

Definition 1

A vine distribution is the distribution of $\mathbf{X} \in \mathbb{R}^m$ with marginal densities f_k , $k \in \{1, \dots, m\}$ and the conditional density of $(X_{j(e)}, X_{k(e)})$ given $\mathbf{X}_{D(e)}$ is specified by $C_{j(e), k(e) | D(e)}$ for the regular vine with the node set $\mathcal{N}$ and the edge set $\mathcal{E}$.

Then, the density of $\mathbf{X}$ is

$$f(\mathbf{x}) = \prod_{k=1}^m f_k(x_k) \prod_{i=1}^{m-1} \prod_{e \in E_i} c_{j(e), k(e) | D(e)}(F(x_{j(e)} | \mathbf{x}_{D(e)}), F(x_{k(e)} | \mathbf{x}_{D(e)}) | \mathbf{x}_{D(e)})$$

Example: in Figure V4, the density of $\mathbf{X}$ is

$$\begin{aligned} f(\mathbf{x}) &= \prod_{k=1}^5 f_k(x_k) \times c_{12} c_{23} c_{24} c_{45} \\ &\times c_{13|2} c_{14|2} c_{25|4} c_{34|12} c_{15|24} c_{35|124}. \end{aligned}$$

All the latter copulas are conditional copulas, but reduces to usual bivariate conditions copulas under the simplifying assumption (SA).

$$\begin{aligned} c_{15|24} &:= c_{15|24}(F(x_1 | x_2, x_4), F(x_5 | x_2, x_4) | x_2, x_4) \\ &\stackrel{SA}{=} c_{15|24}(F(x_1 | x_2, x_4), F(x_5 | x_2, x_4)). \end{aligned}$$

To calculate conditional laws: consider an edge $j, k|D$ in a vine tree.

For any $\ell \in D$ and any $(x_j, \mathbf{x}_D)$, set $D_{-\ell} := D \setminus \{\ell\}$ and

$$F(x_j|\mathbf{x}_D) = \partial_2 C_{j,\ell|D_{-\ell}}(F(x_j|\mathbf{x}_{D_{-\ell}}), F(x_\ell|\mathbf{x}_{D_{-\ell}}); \mathbf{x}_{D_{-\ell}}) \quad (3)$$

In practice: select ℓ so that $j, k|D$ is obtained by connecting $j, \ell|D_{-\ell}$ and $k, \ell|D_{-\ell}$.

⇒ **iterative calculation** of likelihoods, probability of joint events, etc.

This is done in the R-package vine-copula (under (SA)).

Proof of formula (3).

When $\Delta \ll 1$,

$$\begin{aligned} F(x_j | \mathbf{x}_D) &= \mathbb{P}(X_j \leq x_j, X_\ell = x_\ell | \mathbf{x}_{D-\ell}) / \mathbb{P}(X_\ell = x_\ell | \mathbf{x}_{D-\ell}) \\ &\simeq \mathbb{P}(X_j \leq x_j, X_\ell \in [x_\ell, x_\ell + \Delta] | \mathbf{x}_{D-\ell}) / \mathbb{P}(X_\ell \in [x_\ell, x_\ell + \Delta] | \mathbf{x}_{D-\ell}) \\ &= \left\{ C_{j,\ell|D-\ell}(F(x_j | \mathbf{x}_{D-\ell}), F(x_\ell + \Delta | \mathbf{x}_{D-\ell}); \mathbf{x}_{D-\ell}) \right. \\ &\quad \left. - C_{j,\ell|D-\ell}(F(x_j | \mathbf{x}_{D-\ell}), F(x_\ell | \mathbf{x}_{D-\ell}); \mathbf{x}_{D-\ell}) \right\} / \mathbb{P}(X_\ell \in [x_\ell, x_\ell + \Delta] | \mathbf{x}_{D-\ell}) \\ &\simeq \partial_2 C_{j,\ell|D-\ell}(F(x_j | \mathbf{x}_{D-\ell}), F(x_\ell | \mathbf{x}_{D-\ell}); \mathbf{x}_{D-\ell}) \\ &\quad \times \frac{F(x_\ell + \Delta | \mathbf{x}_{D-\ell}) - F(x_\ell | \mathbf{x}_{D-\ell})}{\mathbb{P}(X_\ell \in [x_\ell, x_\ell + \Delta] | \mathbf{x}_{D-\ell})} \\ &= \partial_2 C_{j,\ell|D-\ell}(F(x_j | \mathbf{x}_{D-\ell}), F(x_\ell | \mathbf{x}_{D-\ell}); \mathbf{x}_{D-\ell}) \quad \square \end{aligned}$$

Vines allow highly-flexible models: $m(m-1)/2$ bivariate (possibly conditional) copulas to be specified.

$\Rightarrow$ multiple-stage inference procedures, **tree after tree**.

Imagine we have specified parametric (conditional) bivariate (usual) copulas $C_{j,k|x_D}(\cdot, \cdot; \theta_{jk})$ for all nodes in a regular vine tree.

From an i.i.d. sample $\mathbf{X}_1, \dots, \mathbf{X}_n$, we want to estimate the law of $\mathbf{X}$.

- 1 estimate every marginal law F_k of X_k , $k \in \{1, \dots, m\}$;
- 2 estimate the set of parameters for the nodes of the first tree $T_1 \Rightarrow \hat{\theta}_{j,k}$, $(j, k) \in N_1$.
- 3 use the latter estimated parameters and formula (3) to evaluate the conditional cdfs' $F_j(\cdot | X_k = x_k)$, when $(j, k) \in N_1$ and knowing $\hat{\theta}_{j,k}$.
- 4 evaluate the copula parameters associated to the bivariate copulas in T_2 , using the previously estimated functions $\Rightarrow \hat{\theta}_{jk|l}$ when $jk|l$ is a node of T_2 .
- 5 go on towards the upper tree T_{m-1} .

Example in dimension 3: under (SA), use the decomposition

$$\begin{aligned} f(x_1, x_2, x_3) &= f_1(x_1; \alpha_1) f_2(x_2; \alpha_2) f_3(x_3; \alpha_3) c_{13}(F_1(x_1), F_3(x_3); \theta_{13}) \\ &\times c_{23}(F_2(x_2), F_3(x_3); \theta_{23}) c_{12|3}(F(x_1|x_3), F(x_2|x_3); \theta_{12|3}). \end{aligned}$$

Step 1: estimate the parameters of the margins.

For instance, for any $k \in \{1, 2, 3\}$,

$$\hat{\alpha}_k := \arg \max \sum_{i=1}^n \ln f_k(X_{ik}; \alpha_k).$$

Step 2: estimate the parameters of the copulas on T_1 .

Build pseudo-observations $\hat{U}_{ik} := F_k(X_{ik}; \hat{\alpha}_k)$ for $k \in \{1, 2, 3\}$ and $i \in \{1, \dots, n\}$.

For instance, choose

$$\hat{\theta}_{13} := \arg \max_{\theta_{13}} \sum_{i=1}^n \ln c_{13}(\hat{U}_{i1}, \hat{U}_{i3}; \theta_{13}), \text{ and}$$

$$\hat{\theta}_{23} := \arg \max_{\theta_{23}} \sum_{i=1}^n \ln c_{23}(\hat{U}_{i2}, \hat{U}_{i3}; \theta_{23}).$$

Step 3: estimate the parameters of the conditional copulas on T_2 .

$$\begin{aligned}\hat{U}_{i,1|3} &:= F_{1|3}(X_{i1}|X_{i3}; \hat{\alpha}_1, \hat{\alpha}_3, \hat{\theta}_{13}) \\ &= \partial_2 C_{13}(\hat{U}_{i1}, \hat{U}_{i3}; \hat{\theta}_{13})\end{aligned}$$

$$\begin{aligned}\hat{U}_{i,2|3} &:= F_{2|3}(X_{i2}|X_{i3}; \hat{\alpha}_2, \hat{\alpha}_3, \hat{\theta}_{23}) \\ &= \partial_2 C_{23}(\hat{U}_{i2}, \hat{U}_{i3}; \hat{\theta}_{23})\end{aligned}$$

$$\hat{\theta}_{12|3} := \arg \max_{\theta_{12|3}} \sum_{i=1}^n \ln c_{12|3}(\hat{U}_{i1|3}, \hat{U}_{i2|3}; \theta_{12|3})$$

Without (SA), replace $\theta_{12|3} \in \mathbb{R}$ by a map $x_3 \mapsto \theta_{12|3}(x_3)$.

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